

Local IMS data

Read extracted run-to-failure vibration folders from data/raw.



Feature extraction

Compute time-domain and envelope fault-frequency energy features.



Preprocessing

Normalize features by each run's early healthy baseline and build RUL labels.



Health indicator

Fit PCA-HI and compare it with normalized damage.



Model comparison

Train data-only, physics-informed, LSTM, and CNN RUL models.



Thesis outputs

Write reproducible tables, figures, manifests, and LaTeX assets.